

DECENTRALIZED DIGITAL CURRENCY PROTOCOL**DDCP Architecture — Technical Overview**

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Executive Summary

The Decentralized Digital Currency Protocol (DDCP) is shared, open-source blockchain infrastructure enabling independent operators — fintech companies, cooperatives, exchanges, and civic organizations — to deploy local-currency stablecoins in any jurisdiction where doing so is not prohibited. The protocol requires regulatory permissibility, not government endorsement or co-investment. It delivers fast, cheap, 24/7 payments and financial inclusion using local currency as the unit of account, with tiered privacy and seizure/freeze resistance as protocol-level architectural properties rather than policy settings.

DDCP is explicitly not a CBDC and not a centralized stablecoin. It is the private-sector alternative to both. CBDCs grant governments unprecedented surveillance and control over individual finances; DDCP's architecture structurally prevents that outcome. Centralized stablecoins such as USDC and USDT offer no meaningful privacy and provide issuer-controlled freeze capabilities — as the March 2026 Circle episode demonstrated, a protocol-level freeze function cannot limit its own blast radius and can be compelled by any jurisdiction that can reach the issuer. DDCP's no-freeze and tiered privacy properties are architectural, not contractual: they cannot be revoked by the issuer, compelled away by a single government, or degraded by a governance vote.

The architecture evolves across three stages. Stage 1 deploys on Solana, enabling rapid deployment with sub-second finality and access to an established DeFi ecosystem, within a defined and accepted sovereignty boundary that Stage 2 resolves. Stage 2 transitions to a DDCP sovereign chain with full validator jurisdiction control and native cryptographic privacy enforcement. Stage 3 delivers a purpose-built consensus layer capable of Visa-scale throughput (24,000 TPS) with thousands of politically adversarial validators globally. No specific technology is settled at any

stage beyond Stage 1 — each layer has defined candidates evaluated against explicit criteria.

Core finding: No single existing platform satisfies DDCP's combined requirements of native cryptographic privacy enforcement, protocol-level no-seizure/freeze design, adversarially distributed reserve custody, local-currency multi-instance deployment, and politically independent governance. DDCP occupies uncontested design space at the intersection of these properties. Its architecture leverages Solana at Stage 1 where sovereignty is temporarily and explicitly bounded, and builds natively from Stage 2 where sovereignty is non-negotiable.

1. Fixed Protocol Parameters

The following decisions are fixed protocol parameters. Technology choices within these parameters remain open for evaluation.

Financial Intermediary (FI): Throughout this document, a Financial Intermediary is any entity that (a) crosses the fiat boundary — converting between fiat currency and DDCP tokens in either direction — or (b) holds key custody of private keys authorizing transactions from a user's wallet. The test is functional: an entity's label is irrelevant; what matters is whether either trigger condition is met. The full actor-by-actor classification is in §8.

- **Privacy tiering:** Privacy is tiered based on aggregated wallet transaction value over a rolling protocol-defined period. Retail tier (Tier 1) transactions are cryptographically private and cash-equivalent; large-value tier (Tier 2) is pseudonymous and court-order auditable. Fiat boundary identification is a Financial Intermediary regulatory obligation, not a protocol property. Thresholds evolve through governance within defined bounds and adjust programmatically against inflation over time.
- **AML/CFT at the Financial Intermediary layer only:** AML and CFT compliance is handled entirely at the Financial Intermediary layer. The protocol provides the technical infrastructure for full compliance;

the compliance obligation rests with regulated entities at the fiat boundary.

- **No protocol-level seizure or freeze function:** No protocol-level seizure or freeze operation exists at any layer. This is the affirmatively correct architectural outcome, not a gap. The structural incapacity is the design.
- **Adversarially distributed validators:** Validators are internationally distributed across politically adversarial jurisdictions such that no single government controls a decisive majority.
- **Single-chain architecture:** All DDCP token instances share a single chain, validator network, privacy infrastructure, and consensus mechanism.
- **Gas fees in transaction currency:** Gas fees are paid in the currency used for initiating the transaction.
- **Narrow protocol scope and credit access:** DDCP is narrow infrastructure. The protocol executes value transfer with tiered privacy only. No general-purpose, user-deployable smart contract execution is supported at any stage. DDCP does not host on-chain credit infrastructure. DDUSD and DDEUR holders access existing Ethereum and Solana DeFi via opt-in bridges with documented tradeoffs; local currency token holders access credit through local FinTech lenders at the Financial Intermediary layer.
- **Protocol-level change scope and threshold:** Protocol-level changes are limited in scope and require a high-threshold supermajority of validators, to be predetermined.
- **No protocol-level yield distribution to token holders:** The protocol has no built-in yield distribution mechanism to token holders. Yield flows through the RMF waterfall to Operators. Whether an Operator structures a token holder incentive program or yield passthrough is a deployment-level commercial and regulatory determination. DDUSD specifically excludes yield passthrough due to GENIUS Act constraints; other deployments determine this independently.

2. Design Philosophy

2.1 Financial Privacy and the Procedural Safeguard

DDCP exists because the digitization of money has created a surveillance and control capability with no historical precedent. When money was physical, confiscating it required physical presence, judicial process, and human witnesses. When money is digital and centralized, any entity with administrative access to the payment infrastructure can freeze, block, or redirect funds belonging to any citizen — instantly, without a court order, without notice, and without appeal.

This is not a theoretical concern. In February 2022, the Canadian government invoked emergency powers to freeze the bank accounts of citizens who participated in a lawful political protest — without individual court orders, without criminal charges, and on the basis of political participation rather than criminal conduct. The freezes were executed by administrative directive to financial institutions, not by judicial process, and lifted when the political situation changed — not when courts reviewed them.

DDCP's design goal is to reproduce in the digital realm the essential freedom properties of physical cash. DDCP's tiered privacy architecture reproduces the practical AML effect of cash constraints: retail tier (Tier 1) transactions are cash-equivalent in size and privacy; larger transactions move into pseudonymous and identity-linked tiers with appropriate Financial Intermediary reporting obligations.

DDCP's position rests on a precise distinction: the problem is not that law enforcement should be unable to reach criminal funds. The problem is that centralized digital infrastructure makes politically motivated freezes and legitimate law enforcement freezes structurally identical and equally effortless. DDCP resolves this not by making criminal funds permanently unreachable, but by requiring genuine legal process to reach them. Wallet-level seizure is impossible at the protocol layer. Access restriction through the Financial Intermediary layer remains fully available to law enforcement acting under valid judicial authority.

Financial Intermediaries hold the tools and information AML/CFT compliance requires. FIs maintain KYC identity records linking real-world identities to wallet addresses. FIs are already subject to BSA, FinCEN, and OFAC obligations as licensed financial institutions. Tier 2's

court-order-auditable transaction graph provides the on-chain audit trail for large-value transactions. Critically, the FI layer is the only layer at which the precision law enforcement requires is architecturally achievable: as the March 2026 Circle episode demonstrated, a protocol-level freeze function operates on addresses, not on persons; when an address corresponds to shared infrastructure — a smart contract, a custodial pool, a DeFi protocol — freezing the address freezes every user of that infrastructure regardless of their relationship to the litigation. The Financial Intermediary layer resolves this because FIs hold customer identity records that link legal persons to individual accounts, making targeted enforcement architecturally achievable. AML/CFT compliance is not weakened by locating it at the FI layer; it is strengthened, because that is the only layer where it can work as intended.

2.2 The Validator Censorship Chokepoint

A less visible but equally important dimension of DDCP's design concerns validator governance. The threat is not only that a government will directly order the protocol to implement a freeze. It is that a government will identify the entity that controls validator admission rules and compel that entity to enforce censorship indirectly.

This threat is demonstrated. Following OFAC's 2022 Tornado Cash designation, a majority of Ethereum validators began filtering sanctioned addresses at the block-building level within weeks — not because they were ordered to by a court, but because entities within US jurisdiction faced secondary sanctions exposure. The censorship did not require a protocol-level freeze function.

DDCP's architecture is designed so that no single reachable entity occupies the validator censorship chokepoint. Compelling a decisive majority requires simultaneously reaching entities across multiple non-cooperating legal systems. This requires genuine adversarial distribution — not geographic diversity alone, but structural distribution across jurisdictions with divergent political relationships.

The adversarial jurisdiction distribution of the validator set is the mechanism by which the censorship chokepoint is eliminated at the governance layer, complementing freeze resistance at the protocol layer.

2.3 The No-Freeze Architecture

First: No authority of any kind can freeze a self-custody wallet at the protocol level. The protocol contains no freeze function. This is structural incapacity, not a policy choice.

Second: Judicial orders have a genuine compliance path through the Financial Intermediary layer. A court can order a licensed exchange to restrict a named individual's access to on-ramp/off-ramp services or fiat conversion. Combined with Tier 2's court-order-auditable transaction graph, this provides law enforcement with effective, targeted access — with blast radius naturally limited to the FI-customer relationship.

Third: Administrative directives have no compliance path at any layer for self-custody wallets. The distinction enforced is between enforcement that requires judicial process and enforcement that requires only executive will.

2.4 Architecture in Practice: Recent Demonstrations (2026)

Circle/USDC — March 23, 2026: The blast radius problem.

On March 23, 2026, Circle froze USDC balances in sixteen unrelated operational wallets in response to a sealed civil court order in a private commercial dispute — not a criminal investigation, not a national security matter. Among the sixteen frozen addresses was the ckETH Minter smart contract operated by the DFINITY Foundation — public blockchain infrastructure processing transactions for thousands of users with no relationship to the litigation — and a fully licensed and compliant exchange. Five wallets were eventually unfrozen after public pressure; the remainder stayed frozen.

The problem was not that Circle complied with a court order. The problem is architectural: a protocol-level freeze function cannot limit its own blast radius because it operates at a layer that holds no identity data. DDCP's no-seizure architecture addresses this by ensuring

enforcement actions are executed at the Financial Intermediary layer, where precision is architecturally achievable.

Arbitrum Security Council — April 21, 2026: The justification indistinguishability problem.

On April 21, 2026, Arbitrum’s Security Council exercised emergency governance powers to freeze approximately \$71 million in ETH linked to a bridge exploit attributed to North Korean state-sponsored actors. The outcome was defensible. But the community response identified the structural issue: the action “sets a precedent where with good justification any assets on Arbitrum can be taken from your wallet.”

The architecture cannot distinguish between compelling justifications and pretextual ones. Freeze capability, once embedded in a protocol’s governance layer, is exercised by whoever controls that governance layer, on whatever theory that controller finds sufficient. DDCP’s structural incapacity eliminates both outcomes. The law enforcement interdiction that Arbitrum achieved through emergency governance is available in DDCP’s architecture through the Financial Intermediary layer, under valid judicial authority.

3. Product Architecture

3.1 DDCP Stablecoins — A Better Money

DDCP Stablecoins are pegged 1:1 to the currency of the deployment jurisdiction. Any operator can deploy in any jurisdiction where doing so is not prohibited — no banking license and no government endorsement required. Every DDCP Stablecoin delivers fast and cheap payments available 24/7, local currency denomination, and the unconditional properties — tiered privacy and freedom from arbitrary seizure — that no CBDC and no centralized stablecoin can offer as protocol-level architectural properties.

4. Recommended Architecture

4.1 Architecture Overview — Five-Layer Stack

The recommended architecture is a five-layer stack. No specific technology is settled at any layer — named candidates are alternatives being evaluated against explicit criteria.

Layer naming note: L0 through L4 are DDCP-internal layer designations for this five-layer architecture. They should not be read as references to Layer 0, Layer 1, or Layer 2 in standard blockchain industry vocabulary, where those terms carry specific meanings in the Ethereum scaling ecosystem. In this document, L0 is the reserve layer and L3 is the consensus layer.

Layer	Name	Function / Purpose	Technology Approach	Key Non-Negotiable Properties
L0	Off-chain reserve infrastructure	Hold reserve assets in true custody across adversarially distributed jurisdictions. Orphaned bankruptcy-remote SPVs as the legal isolation structure.	Qualified custodians across adversarially selected jurisdictions (U.S., Switzerland, Singapore, UAE). Permanent partner function — never Build.	True custody — not deposits Zero fractional amplification No single jurisdiction controls a decisive share Proof-of-reserve on-chain attestation Ring-fenced per deployment
L1	Reserve issuance layer	1:1 collateralized token issuance against reserve basket. Coordinates with Financial Intermediaries for mint/burn authorization and AML/CFT compliance. Basket composition and eligible reserve assets defined at RMF governance level.	Evaluated for compatibility with DDCP's reserve composition and governance requirements; built natively if no suitable candidate exists.	1:1 collateralization enforced on-chain Governance-enforced basket composition No holder-level seizure/freeze at any layer
L2	Cryptographic privacy enforcement layer	Enforce tiered privacy at the transaction level. Retail tier (Tier 1): cryptographically	Stage 1: most mature available privacy primitive on the host chain. Stage 2+: DDCP-	Tier 1 boundary: cryptographically enforced, not wallet-self-reported (Stage 2)

		private, enforced by proof circuit. Large-value tier (Tier 2): pseudonymous and court-order auditable.	owned sovereign cryptographic privacy enforcement layer with full Tier 1 activation. Technology selection subject to consumer hardware performance benchmarking.	Cross-token aggregate threshold enforcement (Stage 2) Sub-3s proof generation on mid-tier consumer hardware
L3	Payment chain / consensus layer	Consensus mechanism, validator network, transaction throughput. Single-chain architecture shared across all token instances.	Stage 1: Solana. Stage 2: DDCP sovereign chain with configurable consensus. Stage 3: purpose-built consensus protocol with linear validator communication complexity.	Single-chain architecture High-threshold supermajority for protocol changes No single-jurisdiction validator majority 24,000 sustained TPS target (Stage 3) Full validator jurisdiction control (Stage 2+)
L4	User-facing / application layer	Self-custody wallets, fiat on/off-ramp, DeFi credit access. Default self-custody. Cryptographic proof generation client-side.	Wallet: DDCP-built native wallet + third-party wallets. On/off-ramp: partner providers. DeFi credit: Solana-native protocols for DDUSD & DDEUR at Stage 1; bridge to Solana & Ethereum DeFi at Stage 2+.	Default self-custody No wallet-level seizure/freeze Client-side proof generation Gas fees in transaction currency

4.2 Stage Progression and Technology Evaluation Framework

The architecture evolves across three stages. Stage 1 deploys on Solana — the unconditional starting point, with a defined and accepted sovereignty compromise. Stage 2 delivers full validator sovereignty, native cryptographic privacy, and Tier 1 activation. Stage 3 delivers purpose-built

consensus at Visa-scale throughput. Migration triggers are sovereignty and throughput signals, not calendar dates.

Layer	Stage 1 — Rapid Deployment	Stage 2 — Sovereign Chain	Stage 3 — Purpose-Built Consensus
L0 Reserve	U.S.-qualified custodian at Stage 1. Multi-jurisdiction adversarial custody is the Stage 2 entry condition.	Multi-jurisdiction adversarial custody activated: U.S., Switzerland, Singapore, UAE.	Same structure — reserve custody architecture never changes.
L1 Issuance	Evaluated for compatibility with DDCP reserve and governance requirements; built natively if no suitable candidate.	Fork/Extend → own deployment. Governance outcome applied.	Build if needed. DDCP-owned governance deployed natively if no suitable upstream candidate.
L2 Privacy	Pseudonymous large-value tier only. Cryptographic privacy deliberately held until Stage 2.	Build. DDCP-owned cryptographic privacy enforcement. Tier 1 activated with cryptographic boundary enforcement and cross-token aggregate threshold enforcement.	Full DDCP-native cryptographic privacy. Cross-token aggregate threshold enforcement at full deployment scale.
L3 Consensus	Solana. High-performance, sub-second finality. Defined and accepted sovereignty compromise.	DDCP sovereign chain. Full validator jurisdiction control from Stage 2 day one. Genesis validator set selected per adversarial distribution criteria.	Purpose-built consensus protocol with linear validator communication complexity. Migration trigger: throughput limit reached.
L4 User-facing	DDCP native wallet. On/off-ramp: partner providers. DeFi credit via Solana ecosystem for DDUSD/DDEUR.	Wallet upgraded with Tier 1 cryptographic privacy. Bridge to Ethereum DeFi available for DDUSD/DDEUR (opt-in). On/off-ramp partners maintained.	Full self-custody wallet with complete privacy tier support. Ethereum DeFi bridge maintained as ecosystem infrastructure.

Post-quantum migration pathway (Stage 3 evaluation criterion): The consensus mechanism's validator signature scheme must have a

documented migration pathway to a post-quantum-secure alternative without requiring a validator set replacement or a hard fork that discards accumulated chain state.

4.3 Migration Triggers

Stage 1 → Stage 2: Sovereignty trigger — the point at which Solana’s US regulatory exposure begins to represent a meaningful protocol-level risk given DDCP’s adoption scale. Note: operating two or more DDCP token instances simultaneously at Stage 1 creates a gap in cross-token aggregate threshold enforcement, a Stage 2 cryptographic requirement. The sequencing of a second token deployment should be evaluated against this limitation.

Stage 2 → Stage 3: Throughput trigger — sustained transaction volume approaches Stage 2’s sovereign chain throughput ceiling. Stage 3 build begins in parallel before Stage 2 reaches this limit.

5. Reserve Management Architecture

5.1 Structural Principles

- **Centralized governance, ring-fenced assets:** The RMF centralizes the management function, but each deployment’s reserve assets are independently ring-fenced and independently attested.
- **Orphaned bankruptcy-remote SPVs:** Reserve assets are held through orphaned bankruptcy-remote SPVs legally isolated from any parent entity.
- **True custody, not deposits:** Custodians earn a fee rather than a spread and cannot deploy reserve assets as loanable funds.

5.2 Issuance Control: Mint and Burn Authorization

The mint and burn cycle operates under three co-required controls: (1) the mint function requires co-signatures from both the Issuer and the RMF; (2) a valid current RMF reserve attestation is a protocol-level prerequisite — a mint that would cause circulating supply to exceed attested reserves cannot execute; (3) the RMF will not release reserve assets until the corresponding on-chain burn is confirmed. These controls are symmetric and architectural, not policy settings.

Reserve companion document: For full reserve asset composition, yield waterfall mechanics, stage-by-stage reserve configuration, insurance reserve sizing, and quantitative analysis, see the DDCP Reserve Management Design Specification working paper.

5.3 RMF Governance Requirements

- **Legal structure:** Foundation under neutral jurisdiction (Switzerland for initial formation; Singapore as Stage 2+ candidate), not a corporate entity subject to shareholder control.
- **Authorization:** Multi-signature structure requiring approval from representatives across multiple politically adversarial jurisdictions for any material reserve action.
- **Transparency:** Full on-chain proof-of-reserve attestation at defined intervals. All rebalancing transactions publicly observable.
- **Scope:** Reserve management and yield distribution only. No authority over protocol parameters, validator selection, or token instance governance.

6. Governance Architecture

DDCP has two distinct governance mechanisms operating on entirely different subjects with different decision authorities.

Protocol Governance: The validator-based supermajority process governing protocol-level parameters, the scope of permissible change, and

the properties locked against any change. Operates on the protocol. Has no authority over reserve management.

RMF Governance: The independent foundation process governing reserve management, yield distribution, reserve composition parameters, and deployment-level reserve decisions. Operates on the reserve. Has no authority over protocol parameters or validator selection.

6.1 Protocol Governance — Principles and Locked Properties

Protocol-level changes are limited in scope and require a high-threshold supermajority of validators.

Governance-locked properties: The following cannot be modified by any governance proposal regardless of supermajority size: (1) the retail tier (Tier 1) privacy category — cannot be abolished, collapsed into Tier 2, or redefined below cash-equivalent privacy; (2) cryptographic tier boundary enforcement — governance cannot convert cryptographic enforcement to advisory or client-reported enforcement; (3) universality of the privacy threshold — per-deployment differentiated thresholds are not permissible; (4) the absolute threshold minimum floor and maximum rolling period length ceiling — the threshold cannot be set below a programmatically defined minimum that updates for inflation, and the rolling period length cannot be set above a defined maximum; (5) the no-freeze property — no governance proposal can introduce any protocol-level freeze, seizure, or spending restriction function; (6) the address blacklist prohibition — no governance proposal can require validators to filter transactions from designated addresses.

Governed properties: Within the permissible scope, validators can modify the privacy threshold level, rolling window period, and cross-token aggregate threshold enforcement mechanism, subject to supermajority vote within defined bounds. “Within defined bounds” means two things simultaneously: first, the proposed change must fall within the permissible scope of Protocol Governance and cannot touch any governance-locked property regardless of supermajority support; second, any modified

parameter value must itself fall within the defined ranges — a supermajority can raise the privacy threshold but cannot set it below the locked minimum floor. Reserve basket composition parameters, validator admission criteria, and fee structures are also within governance scope subject to the same two constraints.

Capture resistance: The governance design provides structural capture resistance through a layered structure that allows the validator community to reset governance distributions if active participants appear to have been captured by regulatory pressure, concentrated ownership, or coordinated compulsion.

6.2 Validator Governance — Censorship Chokepoint Elimination

- **No single entity controls validator admission:** Validator admission and expulsion rules must be distributed across participants in politically divergent jurisdictions.
- **Adversarial jurisdiction distribution:** The validator set must be distributed so that compelling a decisive majority requires simultaneously reaching entities across multiple non-cooperating legal systems. A per-jurisdiction concentration ceiling is subject to governance ratification as the genesis validator admission criterion.
- **Address blacklist prohibition:** The protocol must not implement a validator-level address blacklist function. Any governance proposal to introduce address filtering requirements is outside the permissible scope.

The Tornado Cash precedent (2022): Following OFAC’s designation, a majority of Ethereum validators began filtering sanctioned addresses within weeks — not by court order, but because entities within US jurisdiction faced secondary sanctions exposure. DDCP’s adversarial validator distribution and address-blacklist prohibition are the direct architectural responses. The censorship was achieved through the

governance layer — which is why the governance layer must be the locus of the defense.

6.3 Blacklist and Censorship Circumvention — Two Scenarios

Scenario A — Wallet-Level Address Blacklisting (user-facing / application layer — L4): A wallet application maintains an internal address blacklist and refuses to construct or sign transactions involving blacklisted addresses. Governance response: the wallet certification standard explicitly prohibits wallet-level address-based transaction restrictions. Users blocked by a non-certified wallet retain full protocol-layer access through any non-blacklisting wallet, including the open-source reference wallet.

Scenario B — Validator-Level Censorship (consensus layer — L3): Validators filter or exclude transactions from certain addresses at the block-building layer without any protocol change — the demonstrated Tornado Cash mechanism. Governance response: the adversarial validator distribution architecture described in §6.2. The address blacklist prohibition is a bounded-scope protocol parameter that cannot be changed by any proposal within the permitted change scope.

7. Privacy Architecture — Threat-Resistant by Design

The on-chain privacy trilemma — the engineering impossibility of simultaneously achieving usability, absolute privacy, and threat-resistance — establishes that the viable path for financial privacy infrastructure is threat-resistant privacy: full privacy for ordinary activity, combined with cryptographically constrained, narrowly scoped recourse mechanisms for exceptional events. Threat-resistance privacy is an engineering property achieved through architectural choices: tiered disclosure design, cryptographic enforcement of transaction thresholds, and adversarially distributed consensus governance.

DDCP does not claim absolute privacy. It claims threat-resistant privacy — a materially different and architecturally defensible position. This framing is also the affirmative answer to the “absolute privacy enables criminals” objection: DDCP’s architecture resolves the trilemma through tiered design rather than pretending it does not exist. The tiered architecture is also the tool through which Financial Intermediaries meet their BSA/AML obligations: Tier 2’s court-order-auditable transaction graph provides the on-chain audit trail for large-value activity; Tier 1 sits below the cash-equivalent reporting threshold and is handled at the protocol level the same way small cash transactions are treated. The architecture defines the threshold below which compliance is not required and provides the necessary tools above it.

7.1 Two-Dimensional Tier Classification

The tiering system operates on two independent dimensions that must not be conflated or presented as a single graduated scale.

Dimension	Classification Basis	Tiers
On-chain transactions	Aggregated wallet transaction value over a rolling protocol-defined period	Retail tier (Tier 1): below per-token threshold — sender identity, recipient identity, and amount cryptographically concealed; no disclosure path at the protocol layer under any legal process. Large-value tier (Tier 2): above per-token threshold — pseudonymous and court-order auditable.
Fiat boundary crossings	Type of activity — does it cross the fiat boundary or involve key custody?	Identity-linked where required by the Financial Intermediary’s applicable regulatory obligations. Fiat boundary identification is an FI regulatory obligation, not a protocol property.

A \$50 off-ramp processed by a licensed, compliant FI is identity-linked. A \$50,000 on-chain transfer between self-custody wallets is large-value tier (Tier 2). The two dimensions are independent.

Exchange transactions and the threshold: An intra-chain exchange between DDCP token instances (e.g., DDUSD ↔ DDNGN) does not add up to either wallet’s rolling aggregate threshold. The exchange is a

currency denomination change for the same user, not a payment outflow that AML thresholds are designed to monitor. See §9.4 and §12 for full treatment.

Governance-locked privacy properties: (1) Retail tier (Tier 1) privacy category — cannot be abolished, collapsed into Tier 2, or redefined below cash-equivalent privacy; (2) Cryptographic tier boundary enforcement — governance cannot convert cryptographic enforcement to advisory; (3) Universal threshold — per-deployment differentiated thresholds are not permissible; (4) Absolute threshold minimum floor and maximum rolling period ceiling — threshold cannot be set below the defined and programmatically adjusted minimum floor and rolling period cannot be set above the maximum ceiling.

Governed properties: The threshold level, rolling window period, and cross-token aggregate threshold enforcement mechanism are modifiable by supermajority within defined bounds — within permissible scope and within the parameter ranges defined by the locked floor and ceiling.

7.2 Tier 1 — Cryptographic Privacy Enforcement

Tier 1 privacy tier boundaries are enforced cryptographically at the protocol level, not by client self-reporting. Each Tier 1 transaction proof must attest: ‘this transaction is valid AND my rolling aggregate including this transaction remains below the Tier 1 threshold.’ The protocol verifies the proof and rejects the transaction if the proof fails.

Stage 2 design requirement: A wallet-self-reported Tier 1 boundary is trivially defeated by a malicious client and reduces the tiering to a compliance narrative rather than a protocol property. Cryptographic boundary enforcement is a non-negotiable Stage 2 design requirement.

Tier 1 is not activated at Stage 1. Stage 1 launches with Tier 2 pseudonymous transactions only. Tier 1 cryptographic privacy is held until Stage 2, where DDCP owns the cryptographic privacy layer.

7.3 Cross-Token Aggregate Threshold

Per-token threshold enforcement prevents wallets from exceeding the threshold within a single token deployment. Cross-token aggregate threshold enforcement prevents wallets from distributing activity across multiple token deployments to remain below the per-token threshold in each while accumulating above-threshold aggregate value across the protocol.

Formal architecture requirement: The cross-token threshold is the same universal threshold applied to aggregate wallet activity across multiple token deployments simultaneously. Wallet-self-reported enforcement is explicitly rejected. The threshold must be enforced cryptographically at the circuit level.

Two candidate implementation approaches are under active evaluation; technology selection is gated on the consumer hardware performance benchmark.

7.4 Cryptographic Privacy Technology — Evaluation Criteria

No cryptographic privacy implementation is settled. Any candidate must be evaluated against:

- Sub-3-second proof generation on mid-tier consumer Android and iOS hardware for retail tier transaction amounts — with the full proof including transaction validity, per-token aggregate compliance, and cross-token aggregate compliance.
- A documented migration pathway to a post-quantum variant without requiring a full circuit rewrite.

- Cryptographic support for cross-token aggregate threshold enforcement as a cryptographic property, not a self-reported one.
- Circuit design that can be owned and optimized by DDCP directly.
- No dependency on a third-party blockchain platform subject to adverse jurisdiction regulatory exposure.

8. Financial Intermediary Classification

8.1 The Operative Test

A Financial Intermediary is any entity that (a) crosses the fiat boundary — converting between fiat currency and DDCP tokens in either direction — or (b) assumes key custody of private keys authorizing transactions from a user's wallet. The test is functional, not categorical.

There are two distinct levels of fiat boundary crossing:

Level	Entity	Fiat boundary type	Primary obligation
Structural	Issuer (DDUSD, LLC)	Aggregate token issuance and redemption against reserve	Reserve adequacy, PPSI compliance, redemption guarantee
Transactional	On/off-ramp, exchange, custodial wallet	Individual user fiat ↔ token conversion or key custody	KYC/KYB, transaction monitoring, Travel Rule, SAR

8.2 Actor Classification

Actor Type	Financial Intermediary?	Trigger / Notes
Fiat on/off-ramp operator	Yes — Transactional	Fiat boundary trigger. Primary AML/CFT obligation bearer.
Custodial wallet provider	Yes — Transactional	Key custody trigger. Full FI obligations apply.

Non-custodial wallet (keys on user device only)	No	User signs unilaterally. Wallet provider has no key material and crosses no fiat boundary. On-chain swaps between DDCP token instances via a protocol-level swap primitive, if implemented, do not constitute a fiat boundary crossing but can trigger DASP obligations. See §12 for the protocol-level atomic swap primitive design question.
MPC wallet — server share is recovery only	No	User signs unilaterally in normal operation. Server share is a recovery mechanism only.
MPC wallet — both shares required to sign	Yes — Structural	Server is a required co-signer. Key custody trigger.
DDCP Foundation, Inc.	No	Neither trigger condition met. Stewards the open-source protocol specification and administers the DDCP certification mark.
Operator / Issuer	Yes	Fiat boundary trigger at the wholesale issuance layer. AML/CFT obligations are issuer-level. Retail KYC, individual transaction monitoring, Travel Rule compliance, and SAR reporting attach at the exchange and on-ramp layer.
DDCP Reserve Management Foundation (RMF)	No	Neither trigger condition met. Manages reserve assets only; does not interact with user wallets, cross the fiat boundary, or hold user private key material.

Sending-wallet privacy asymmetry: In a transaction between a standard wallet and a DDCP-native wallet, the privacy of the transaction is determined primarily by the sending wallet, not the receiving wallet. The sender bears the proof generation obligation. A non-certified sender can reduce a transaction to Tier 2 even when transacting with a certified recipient. This is an additional argument for wallet certification: a non-certified wallet cannot guarantee Tier 1 privacy for its counterparty. Non-DDCP tokens transacted through a DDCP-native wallet use the non-DDCP token's own infrastructure and have no DDCP privacy guarantee regardless of which wallet is involved. See §12 for open questions on this interaction.

9. Bridge Architecture and On-Chain Credit Access

9.1 DDCP's Role: Promoting External Credit Access

DDCP does not host on-chain credit infrastructure. The protocol provides unfreezable settlement rails for base money; credit ecosystems build on top of or alongside those rails. On-chain credit access must not be used as a gating condition for adoption.

9.2 External Credit Access by Token Type

DDUSD and DDEUR: Holders access the existing Solana DeFi credit ecosystem via opt-in bridge from Stage 1. DDUSD and DDEUR carry no foreign exchange risk relative to USD and EUR DeFi frameworks, so collateralization requirements are comparable to USDC and EURC respectively. Bridge tradeoffs — departure from DDCP's guarantee envelope, exposure to destination-chain freeze vectors — are documented transparently as user-assumed risk.

Local currency tokens: Holders access credit through local FinTech lenders at the Financial Intermediary layer. Solana & Ethereum DeFi is technically accessible via bridge from Stage 2, but is economically impractical for most users due to overcollateralization requirements on FX-volatile collateral.

9.3 Bridge Infrastructure — Scope and Limits

From Stage 2 onward, DDCP specifies a bridge-compatible token interface standard so that third parties can build bridges connecting DDCP tokens to Solana and Ethereum. DDCP does not build, maintain, or endorse any specific bridge implementation.

Bridge risk disclosure: Users who bridge tokens accept an opt-in custodial dependency outside DDCP's no-seizure/freeze guarantee. The mechanism is lock-and-mint: tokens are locked in a bridge contract on

the DDCP chain; a wrapped representation is minted on the destination chain. If the wrapped representation is frozen on Ethereum, the bridge contract on the DDCP chain does not receive the burn signal required to release the locked tokens. The underlying DDCP chain has no freeze function — the user's tokens are nonetheless inaccessible because the bridge layer does.

9.4 Cross-Token Settlement and Threshold Accounting

Cross-token settlement on DDCP — enabling value transfer between DDUSD and DDNGN, for example — is a planned workstream. Because DDCP is single-chain, cross-token settlement between instances is an intra-chain operation: there are no chain hops, and both token instances share the same L2 privacy layer and L3 consensus. Privacy properties for cross-token swaps are enforced by the same single-chain proof circuit.

Architecture evaluation criteria:

- Settlement atomicity and failure mode handling: the architecture must define a rollback or recovery mechanism for partial completion scenarios.
- **Threshold accounting — DDCP-to-DDCP swaps:** An intra-chain exchange between DDCP token instances (DDUSD ↔ DDNGN) does not count toward either wallet's rolling aggregate threshold. The swap is a currency denomination change for the same user, not a payment outflow that AML thresholds are designed to monitor. Circuit design implications require explicit resolution before Stage 2.

Non-DDCP to DDCP swaps and threshold netting: If a protocol-level atomic swap primitive is implemented (see §12), a swap from a non-DDCP token into a DDCP token cannot be netted at the threshold circuit level: the DDCP privacy circuit has no visibility into the non-DDCP side of the transaction and cannot attest a net position. The DDCP value received in such a swap counts toward the receiving wallet's rolling aggregate threshold. This creates a design tension — a large non-DDCP

to DDCP onboarding swap consumes threshold headroom immediately on arrival — that must be addressed in the circuit design specification.

10. Wallet Architecture and Protocol Guarantees

10.1 Protocol vs. Application-Layer Guarantees

Feature	Standard Wallet	DDCP-Native Wallet	Layer
Fast, cheap, 24/7 payments	✓	✓	Protocol
Local currency denomination	✓	✓	Protocol
No protocol-level seizure	✓	✓	Protocol
No token expiration	✓	✓	Protocol
No vendor or goods restrictions	✓	✓	Protocol
Pseudonymous transactions (Tier 2)	✓	✓	Protocol
Unconditioned transactions	✗	✓	Application (L4)
No wallet-level address blacklist	✗	✓	Application (L4)
Cash-equivalent Tier 1 privacy (Stage 2+)	✗	✓	Application (L4)
Full DDCP proposition	✗	✓	Application (L4)
Intra-chain DDCP token swaps (DDUSD ↔ DDNGN)	✓	✓	Protocol (open design question — see §12)

10.2 Circumvention Vectors — Standard Wallets

Vector	Scenario	Description	Guarantee Affected
UI-layer conditioning	Application layer (L4)	A compelled wallet refuses to build transactions to certain addresses, requires additional approvals, or silently modifies transaction parameters before signing.	Unconditioned transactions
Address blacklisting (Scenario A)	Application layer (L4)	A wallet maintains an internal address blacklist and refuses to construct or sign transactions involving blacklisted addresses. See §6.3.	No wallet-level address blacklist
Broadcast layer interdiction	Application layer (L4)	If the wallet routes broadcasts through its own servers, it can drop, delay, or conditionally block transactions without the user's knowledge.	Unconditioned transactions
Privacy layer bypass	Application layer (L4)	A compromised wallet can skip privacy wrapping, generate proofs revealing more than required, or log and exfiltrate transaction details.	Cash-equivalent Tier 1 privacy
Metadata leakage	Application layer (L4)	Even a correctly functioning wallet leaks metadata the protocol does not protect: IP address, timing, device fingerprint, network traffic.	Cash-equivalent Tier 1 privacy (practical)
Validator-level censorship (Scenario B)	Consensus layer (L3)	Validators filter transactions from certain addresses at the block-building layer without any protocol change — the demonstrated Tornado Cash mechanism. See §6.2 and §6.3.	Protocol-level censorship resistance

10.3 Wallet Certification Standard

- **Open-source reference wallet:** DDCP maintains and publishes an open-source reference wallet implementing all privacy tiers, routing broadcasts directly to validators without intermediary servers, with no silent transaction parameter modification, and with no address blacklist.
- **Wallet certification requirements:** Direct validator broadcast; client-side cryptographic proof generation; prohibition on silent transaction

parameter modification; no address blacklist; metadata minimization standards.

- **User-verifiable transaction confirmation:** A protocol-level transaction receipt that users can independently verify was broadcast as signed. This is a certification requirement.

11. Stage Configurations

11.1 Stage 1 — Rapid Deployment

Role: Unconditional starting stage. Fastest path to live deployment and early adoption. No stage-entry conditions.

Stage 1 deploys on Solana as the L3 host chain — high-performance, sub-second finality. Pseudonymous large-value tier (Tier 2) privacy at L2. L1 issuance layer under evaluation. U.S.-qualified custodian at L0. Self-custody wallet at L4 with opt-in access to Solana DeFi for DDUSD/DDEUR.

Dimension	Configuration
Validator sovereignty	Partial — Solana is US-headquartered. Adversarial jurisdiction requirement cannot be enforced at the base layer. This is the defined and accepted sovereignty compromise. Stage 2 resolves it.
Privacy tier active	Tier 2 pseudonymous only. Tier 1 cryptographic privacy deliberately held for Stage 2.
No-seizure at L3	Dependent on Solana network governance. Accepted as a temporary stage constraint.
Reserve custody	U.S.-qualified custodian. Multi-jurisdiction adversarial custody is the Stage 2 entry condition.
DeFi credit — DDUSD & DDEUR	Via opt-in bridge to Solana DeFi ecosystem.
DeFi credit — local currency tokens	Local FinTech lenders at the Financial Intermediary layer.

US regulatory surface	High — Solana is US-headquartered. Migration trigger: Solana regulatory exposure reaches protocol-level risk.
Time to first deployment (estimate)	6–12 months from foundation activities.
Migration trigger to Stage 2	Sovereignty trigger — Solana regulatory exposure represents meaningful protocol-level risk at current adoption scale.

11.2 Stage 2 — Sovereign Chain

Role: Full validator sovereignty. Native cryptographic privacy ownership and Tier 1 activation. Stage 2 entry condition: multi-jurisdiction adversarial custody activated.

Stage 2 architecture: DDCP sovereign chain at L3 with full validator jurisdiction control. DDCP-owned cryptographic privacy at L2 with Tier 1 activation. Multi-jurisdiction adversarial custody at L0. Bridge infrastructure to Solana and Ethereum available as opt-in ecosystem service.

Dimension	Configuration
Validator sovereignty	Full — no dependency on any existing L1's governance. Full validator jurisdiction control from Stage 2 day one.
Privacy tier active	Tier 1 cryptographic privacy activated with cryptographic boundary enforcement and cross-token aggregate threshold enforcement.
No-seizure at L3	Enforced by DDCP protocol.
Reserve custody	Multi-jurisdiction adversarial custody: U.S., Switzerland, Singapore, UAE.
DeFi credit — DDUSD & DDEUR	Bridge to Ethereum DeFi available as opt-in. Solana DeFi also accessible via bridge.
Governance	DDCP Protocol Governance and RMF Governance independently operational.
Migration trigger to Stage 3	Throughput trigger — sustained volume approaches Stage 2's sovereign chain throughput ceiling.

11.3 Stage 3 — Purpose-Built Consensus

Role: All long-term production deployments at scale. Purpose-built BFT consensus with linear validator communication complexity, supporting thousands of politically adversarial validators globally.

Dimension	Configuration
Consensus layer	Purpose-built consensus protocol with linear $O(n)$ validator communication complexity.
Throughput target	24,000 sustained TPS — Visa-scale throughput benchmark.
Validator scale	Thousands of validators globally. Adversarial jurisdiction distribution maintained at scale.
Privacy tier	Full DDCP-native cryptographic privacy. Cross-token aggregate threshold enforcement at full deployment scale.
Reserve custody	Same multi-jurisdiction adversarial custody as Stage 2 — permanent.
Cross-token settlement	Cross-token settlement mechanism between DDCP instances operational.
Governance	Full DDCP Protocol Governance and RMF Governance. Participants distributed across 20+ jurisdictions.

11.4 Stage Comparison

Dimension	Stage 1	Stage 2	Stage 3
Validator sovereignty	Partial — Solana dependency (defined and accepted)	Full	Full
No-seizure at L3	Dependent on Solana governance	Protocol-enforced	Protocol-enforced
Privacy — Tier 1	Not available	Activated (cryptographic enforcement)	Full native implementation

Privacy — Tier 2	Available	Available	Available
Cross-token threshold enforcement	Not available	Activated (Stage 2 entry)	Full implementation
Reserve custody	Single U.S. custodian	Multi-jurisdiction adversarial	Multi-jurisdiction adversarial (permanent)
Throughput benchmark	~65,000 TPS (Solana)	Significant with consensus customization	24,000 TPS target (purpose-built)
DeFi credit — USD/EUR tokens	Solana-native protocols	Bridge to Solana & Ethereum DeFi available	Bridge maintained as infrastructure
Bridge to Solana	Not needed — native	Available (opt-in)	Available (opt-in, maintained)
Bridge to Ethereum	Not available	Available (opt-in)	Available (opt-in, maintained)
Governance capture resistance	Solana governance (limited)	DDCP Protocol Governance active, capture-resistant	Full DDCP Protocol and RMF Governance deployed
US regulatory surface	High — Solana dependency	Minimal	Minimal

12. Open Questions Requiring Resolution

The following questions require dedicated workstreams or governance decisions. None are re-litigations of settled architecture.

Technical

Question	Status / Notes
Cryptographic privacy technology selection and performance benchmark	Performance benchmarking on mid-range consumer Android and iOS hardware for DDCP's specific circuit requirements: transaction validity + per-token aggregate compliance +

	cross-token aggregate compliance. Target: sub-3 seconds. Results inform circuit design and technology selection; do not gate Stage 1 entry.
Cross-token aggregate threshold — implementation approach	Two candidate approaches: shared cross-token state registry vs. cross-circuit proof aggregation. Must meet the sub-3-second consumer hardware target with cross-token aggregate compliance included. Wallet-self-reported enforcement is explicitly rejected.
Protocol-level atomic swap primitive	Whether DDCP should implement a protocol-level atomic swap primitive enabling intra-chain exchange between DDCP token instances (e.g., DDUSD ↔ DDNGN) without a general-purpose smart contract. Open design question given the narrow infrastructure principle: any swap primitive must be a specific, bounded protocol function, not a general-purpose programmable layer. Design implications: (a) FI status of the swap function itself; (b) threshold accounting for DDCP-to-DDCP swaps (working design position: does not count) vs. non-DDCP-to-DDCP swaps (DDCP value received counts; netting is not architecturally available at the circuit level); (c) application-layer disclosure obligations for non-DDCP token privacy properties. Resolution required before any intra-chain exchange capability is advertised. Validate that this approach does not trigger DASP obligations on the Protocol, under GENIUS Act Section 2(7)(A)(ii).
Cross-token settlement mechanism design	Evaluate candidate architectures for value transfer between DDCP token instances. Single-chain architecture means no chain hops; privacy properties are enforced by the shared L2 proof circuit. Settlement atomicity and failure mode handling must be specified.
Wallet security standards and certification criteria	Pre-mainnet requirement. Direct validator broadcast, client-side proof generation, transaction parameter integrity, metadata minimization, no address blacklist. Formal specification required before DDCP-native third-party wallets can be endorsed.
Sending-wallet privacy asymmetry	A transaction between a standard wallet and a DDCP-native wallet has privacy properties determined primarily by the sending wallet's proof capability. A non-certified sender can reduce a transaction to Tier 2 even when transacting with a certified recipient. Implications for: (a) wallet certification standard and user communication; (b) whether the protocol should reject Tier 1 proof-less transactions from standard wallets when the recipient is a certified wallet; (c) disclosure requirements at the application layer.

Non-DDCP token privacy in DDCP-native wallets	Privacy is a property of the DDCP token layer, not the wallet application. A DDCP-native wallet can display and manage non-DDCP tokens, but those transactions use the non-DDCP token's own infrastructure and have no DDCP privacy guarantee. Implications for user disclosure and wallet UI design.
Operative validator country concentration threshold	Per-jurisdiction concentration ceiling under development as genesis validator admission criterion. Subject to governance ratification.
Multi-wallet AML limit circumvention	Cross-token aggregate threshold enforcement and FI-layer aggregate flow monitoring are the primary detection mechanisms. Additional protocol-level response options under evaluation.
Bridge-compatible token interface standard	Stage 2 design requirement. Specification of what a DDCP token must implement to be bridgeable without compromising the base-layer no-seizure guarantee. Governance ratification required.
Oracle infrastructure for local currency tokens	Pre-local-currency-token deployment. Coverage and reliability requirements must be confirmed before local currency token launch.
Threshold reference currency and inflation oracle	Absolute threshold minimum floor referenced to a protocol-designated fiat currency (initially USD / DDUSD) updated programmatically using an inflation oracle (initially US CPI). Reference currency, CPI oracle, and exchange rate oracle are governed parameters. Oracle provider selection and fallback mechanism must be specified before Tier 1 activation.
Intra-chain token swap threshold accounting	Intra-chain DDCP-to-DDCP swaps: working design position is that these do not contribute to the rolling aggregate threshold (denomination change, not a payment outflow). Non-DDCP-to-DDCP swaps: netting is not architecturally available at the circuit level; the DDCP value received counts toward the receiving wallet's threshold. Both positions require circuit design resolution before Stage 2 deployment.
Post-quantum consensus migration specification	Stage 3 consensus technology selection must include confirmation that the chosen signature scheme has a viable post-quantum successor and that the validator key migration protocol is specified before genesis validator commitments are solicited.

Economic

Question	Status / Notes
On-chain credit stratification between token types	DDUSD & DDEUR access existing Solana DeFi via opt-in bridge; local currency tokens access credit through local FinTech at the FI layer. This credit access asymmetry is a strategic deployment sequencing observation, not a DDCP architecture workstream.
Commodity index inclusion — timing and governance	Stage 2+. Governance ratification required before inclusion.
Coordinated global monetary expansion risk	Ongoing monitoring. COVID-era QE across all major reserve issuers is a known design boundary affecting the sovereign debt tranche. Monitoring framework required.
Legal tender recognition as adoption limitation	Open. Whether absence of legal tender status materially limits adoption in target jurisdictions requires per-jurisdiction assessment.

Legal, Regulatory, and Governance

Question	Status / Notes
FATF engagement strategy	FATF guidance on decentralized protocols affects the regulatory permissibility baseline. The Financial Intermediary layer fully satisfies AML/CFT obligations under the DDCP architecture. Engagement recommended as an early priority.
MiCA applicability	MiCA carve-out for fully decentralized protocols may apply but cannot be assumed. Legal assessment required before any EU market entry or DDEUR deployment.
GENIUS Act PPSI license application	DDUSD, LLC is pursuing a Permitted Payment Stablecoin Issuer (PPSI) license under the GENIUS Act with the OCC. Application to be filed after the final implementing rule is published.
Genesis validator selection criteria and candidates	Formal criteria required: jurisdiction concentration ceiling, conflict-of-interest framework, minimum technical requirements, adversarial distribution verification methodology.
Improvement proposal process specification	Submission standards, deliberation periods, and voting thresholds by proposal type. Scope of permissible changes to be bounded and published explicitly.
RMF governance specification	Legal structure, multi-sig composition and decision rules, proof-of-reserve attestation cadence, rebalancing trigger

	criteria, conflict-of-interest framework, and scope boundary. Must be complete before reserve deployment.
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All architectural parameters, stage definitions, and open questions are subject to revision by the DDCP governance process.